arm

Cortex-M85 high performance Cortex-M processor

Edge AI LoB

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Unlocking Use-cases with highest performance Cortex-M

IoT/Embedded

- Smart speakers
- Utility robots
- Drone control

Client

- AR/VR
- Secured system controller
- Sensor hubs

Automotive

- Intelligent actuation
- radar
- EV motor control

Improved Cortex-M CPU capabilities: Market needs addressed

Step up in performance
Local processing, less traffic to cloud
Intelligence at the edge

→ Advanced features and capability
→ Better power efficiency
→ More reliable and faster user interface
→ Enables new applications

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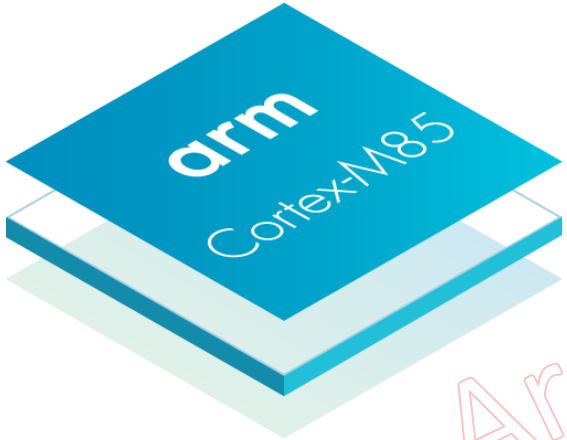
Cortex-M85: Highest Performing Cortex-M Processor Ever

Pushing the boundaries for secure, high-performance microcontrollers



Highest performance Arm MCU ever across both traditional and ML workloads

Enhanced micro-architecture features for increased scalar performance.

>6 CoreMarks/MHz

Helium enables ML and DSP workloads.

4x ML performance vs. Cortex-M7

3x DSP performance vs. Cortex-M7

Enhanced security with Arm TrustZone and PACBTI (Pointer Authentication and Branch Target Identification)

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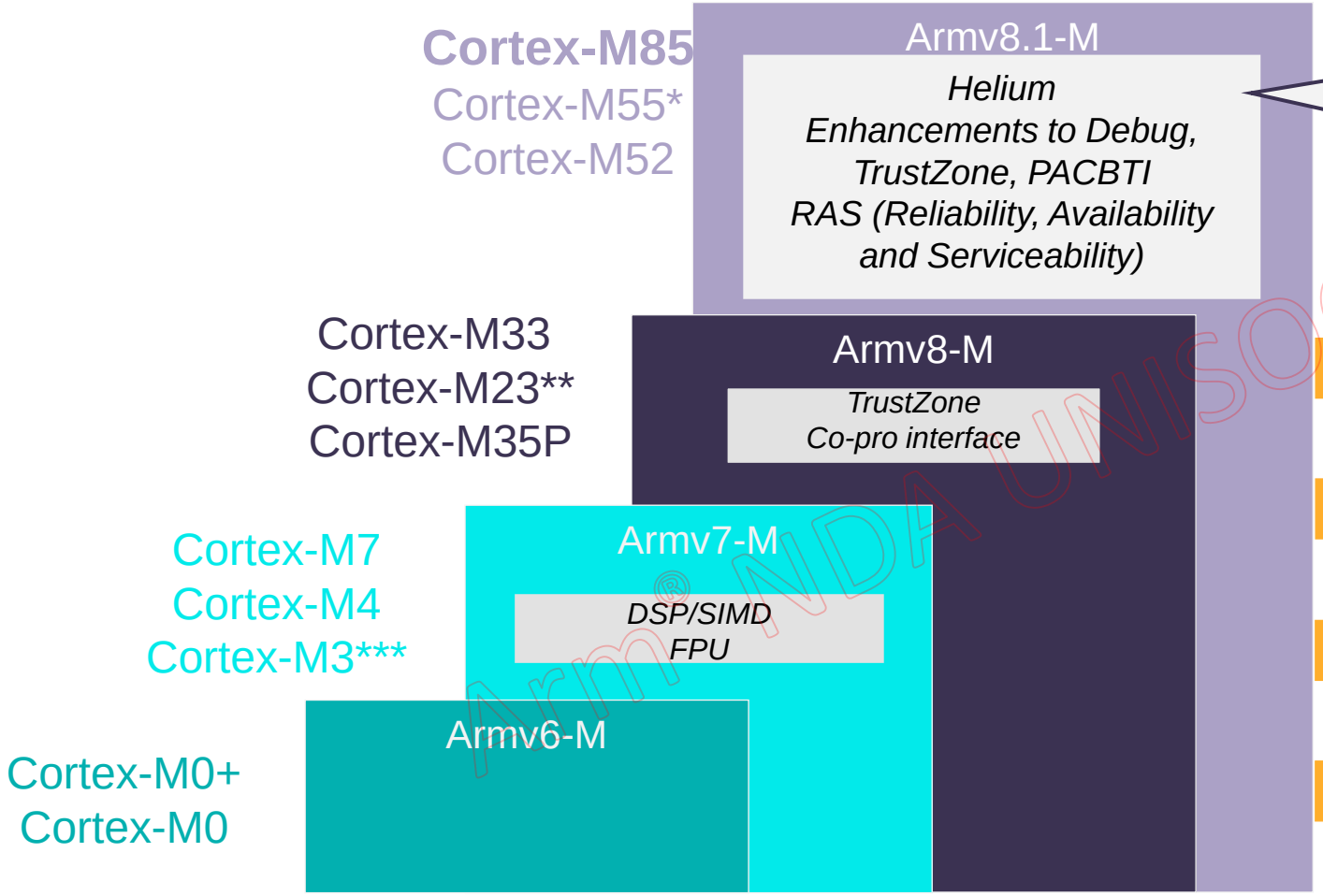
Architecture evolution for smarter and more secure systems



>150 new scalar and vector instruction
Low overhead loops
Predication
Arithmetic support for 8-bit fixed and 16-bit float
Gather load, scatter store
Complex math

Ease of development

Energy-efficient compute

Increased throughput

Smarter devices

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* Cortex-M55 does not include PACBTI

** Cortex-M23 does not include co-processor interface

*** Cortex-M3 does not include DSP/SIMD and FPU

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Cortex-M Portfolio

	Armv6/7-M	Armv8-M *(with TrustZone™)*	Armv8.1-M *(with Helium™)*
High performance	Cortex-M7		**Cortex-M85** • Highest scalar performance • Arm Custom Instructions • Enhanced Functional Safety
Mainstream	Cortex-M4 Cortex-M3	**Cortex-M33** • Mainstream processor with TrustZone	**Cortex-M55** • Arm Custom Instructions • Enhanced Functional Safety **Cortex-M52** • Smallest Helium • Arm Custom Instructions • Enhanced Functional Safety
Constrained	Cortex-M0+ Cortex-M0	**Cortex-M23** • Smallest Safety Processor • Enhanced Functional Safety	

Last Update: May 2024

Released (EAC/REL)

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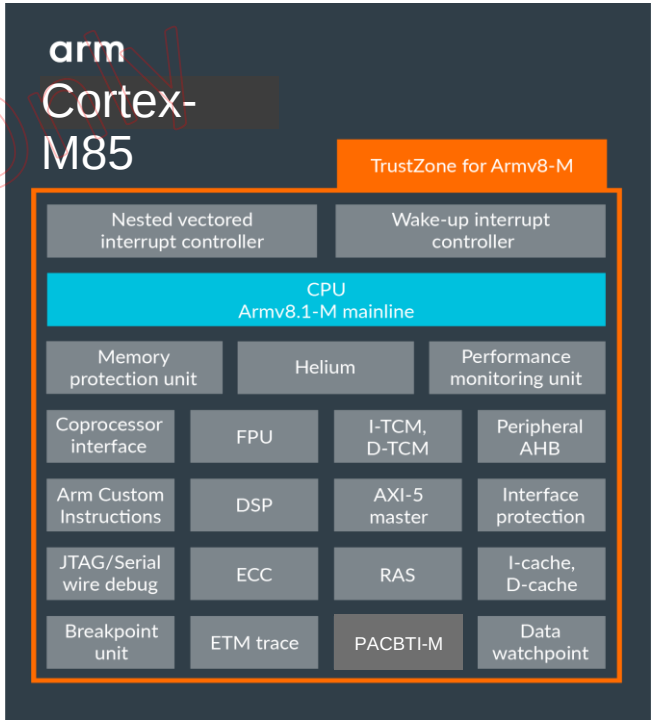
Cortex-M85

Highest performance, next-generation Cortex-M

- unprecedented performance with built-in security
  - 64b instruction fetch, full dual-issue, delivering unprecedented scalar performance
  - dual-beat Helium™ delivers multi-fold vector performance increase
  - extended pipeline to achieve on-par frequencies as Cortex-M7
  - built-in TrustZone, optional PACBTI

- wide choice of options for differentiated products
  - PACBTI-M mitigates SW ROP/JOP attacks
  - Helium and Floating-Point Unit (FPU) individually optional
  - coprocessor interface
  - comprehensive debug and real-time trace (ETM, ITM, DWT)



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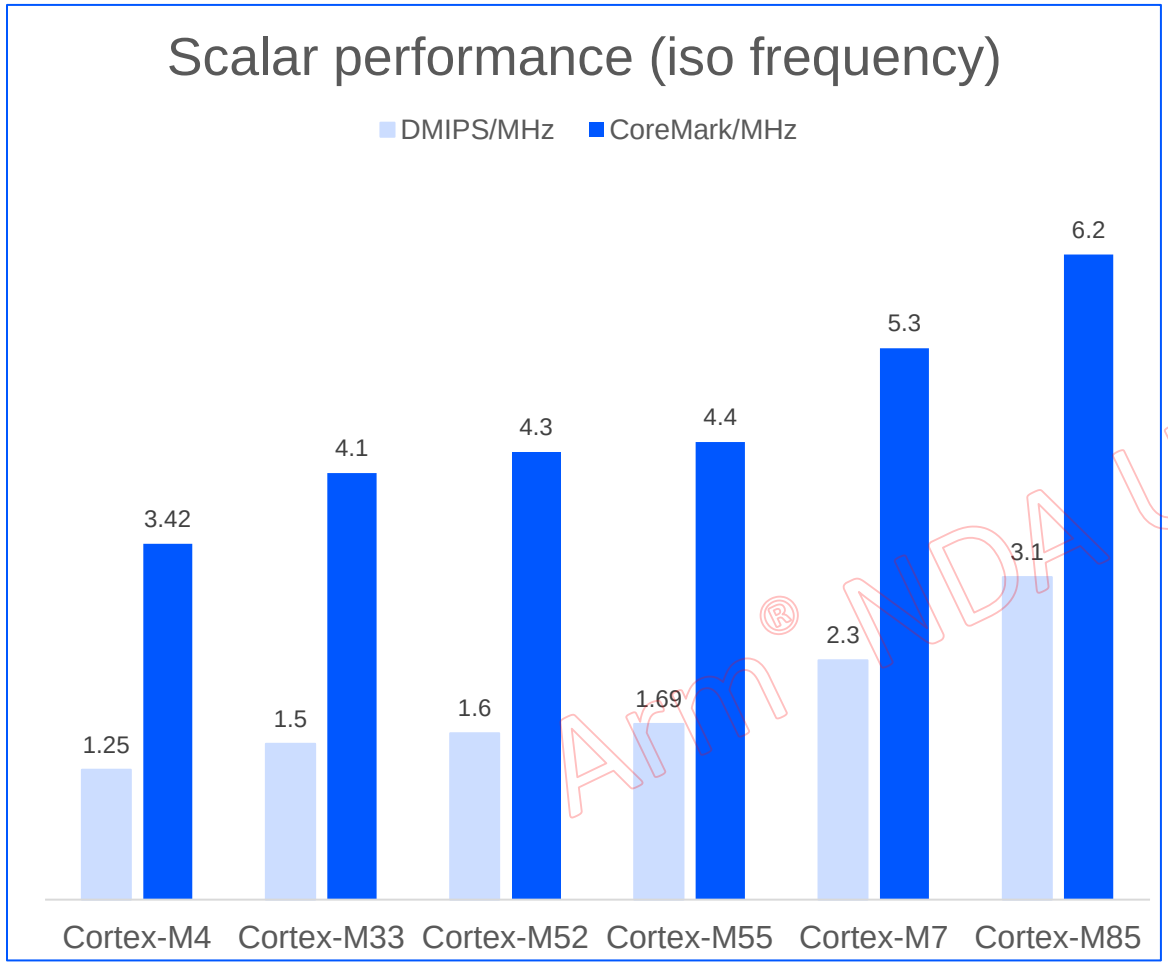
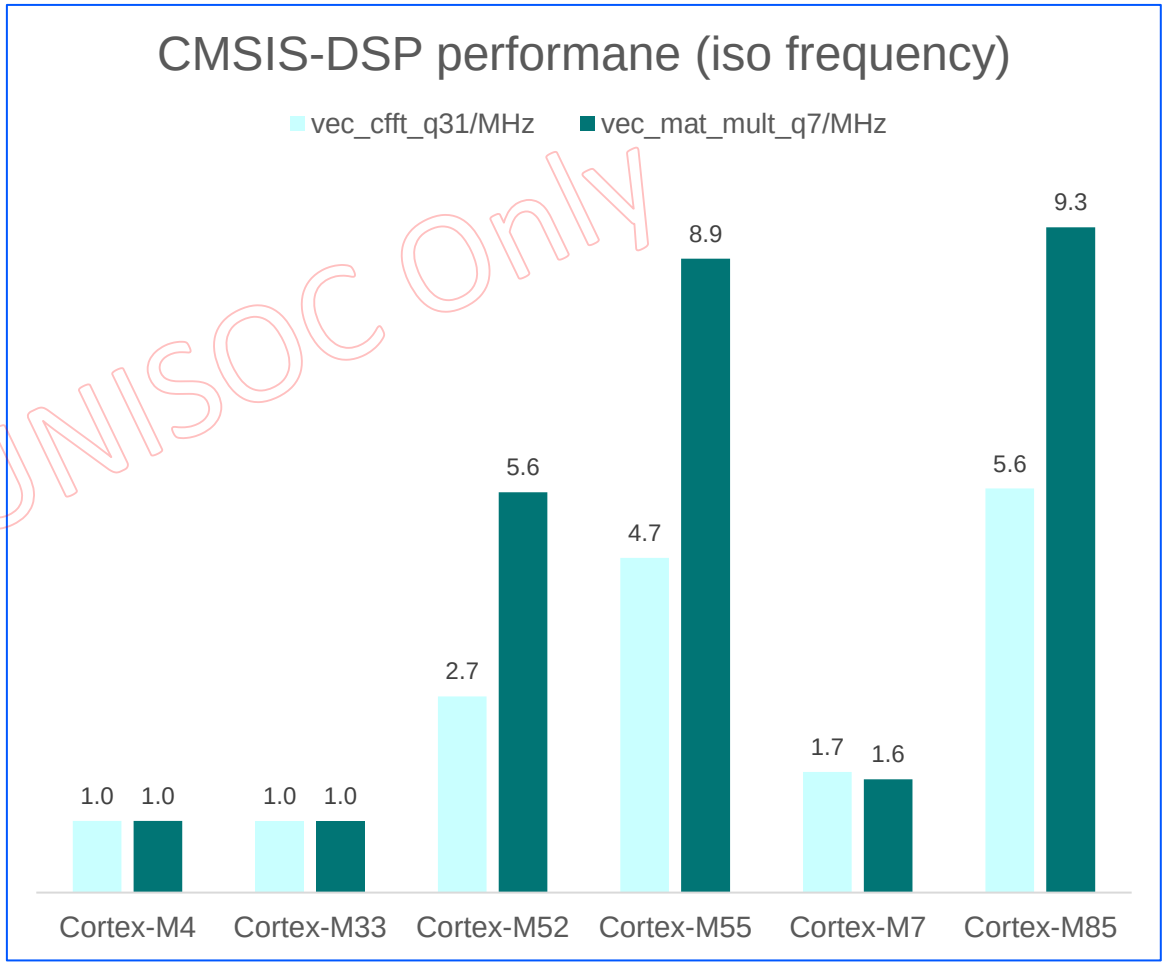
Performance Benchmark targets (iso frequency)





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Cortex-M processor comparison

		Cortex-M33	Cortex-M52	Cortex-M55	Cortex-M7	Cortex-M85
	Instruction set architecture	Armv8-M	Armv8.1-M	Armv8.1-M	Armv7-M	Armv8.1-M
Performance	DMIPS/MHz	1.50	1.6	1.69	2.14	3.1
	CoreMark/MHz	4.02	4.3	4.4	5.3	6.2
	CFFT Q31/MHz (normalized)	1	2.7	4.7	1.7	5.6
	Matmul Q7/MHz (normalized)	1	5.6	8.9	1.6	9.3
	Approximate sustained MACs/cycle	~0.5 32bx32b	1 32bx32b 2 16bx16b 4 8bx8b	2 32bx32b 4 16bx16b 8 8bx8b	~1 32bx32b	2 32bx32b 4 16bx16b 8 8bx8b
Key microarchitecture features	Pipeline stages	3	~4	4	6	7
	Helium (M-profile Vector Extension)	No	Yes (single-beat)	Yes (dual-beat)	No	Yes (dual-beat)
	Floating point hardware	Yes (SP)	Yes (HP, SP, DP)	Yes (HP, SP, DP)	Yes (SP, DP)	Yes (HP, SP, DP)
	Built-in Caches	No	0-64kB I- and D-cache + optional unified cache	0-64kB I- and D-cache	0-64kB I- and D-cache	0-64kB I- and D-cache
	Tightly Coupled Memory	No	0-16MB I- and D-TCM	0-16MB I- and D-TCM	0-16MB I- and D-TCM	0-16MB I- and D-TCM
	TrustZone for Armv8-M	Yes	Yes	Yes	No	Yes
	Maximum MPU regions	16	16	16	16	16
	Co-processor interface	Yes	Yes	Yes	No	Yes
	Arm Custom Instruction	Yes	Yes	Yes	No	Yes
	Main bus interface	AMBA5 AHB	AMBA5 AXI/AHB	AMBA5 AXI	AMBA4 AXI	AMBA5 AXI
	Maximum # external interrupts	480	480	480	240	480
Functional safety features	DCLS support	Application note	Reference implementation	Reference implementation	Reference implementation	Reference implementation
	MBIST	N/A	Yes	Yes	No	Yes
	Bus parity	No	Yes	Yes	No	Yes
	ECC	N/A	Both caches and TCMs	Both caches and TCMs	Caches only	Both caches and TCMs
	Transient fault protection (flop parity)	No	Yes	Yes	No	Yes
Availability		Released	Released	Released	Released	Released

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Fast Track Your Arm Cortex-M85 Physical Implementation

PIK implementation best practices accelerate time-to-market and reduce design risk

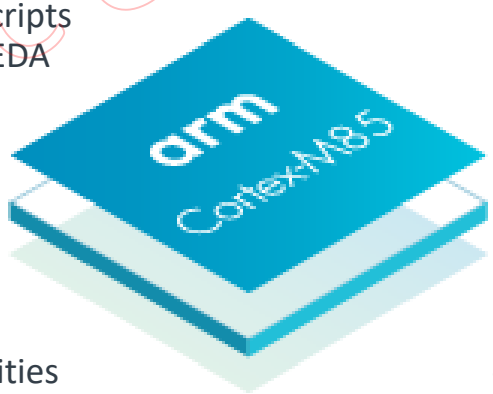
Processor Implementation Kit (PIK)

- Leverages Arm off the shelf physical IP portfolio across foundries and process nodes
- Builds on Arm expertise across RTL, physical
  IP, process and EDA tools for optimal PPA
- Multiple reference flows & Shmoo plots tailored to meet performance, low power, area, cost requirements trade-offs
- Implementation best practices for target Arm M-class processors provides 2 - 4 months TTM advantage
- 1st Cortex-M85 PIK targets TSMC 22nm ULL popular IoT foundry process node

Arm Cortex-M85 PIK for TSMC 22ULL

RTL-GDS scripts for Major EDA tool chains

Comprehensive implementation methodology



Design utilities to improve implementation

Shmoo plots (PPA trade-offs)

TSMC 22ULL PPA trade-offs

- 800 MHz, Balanced PPA*
- 1GHz, Max frequency
- 600Mhz, Optimal low power

* 1st Cortex-M85 PIK release focuses on balanced PPA

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Corstone for Cortex-M85

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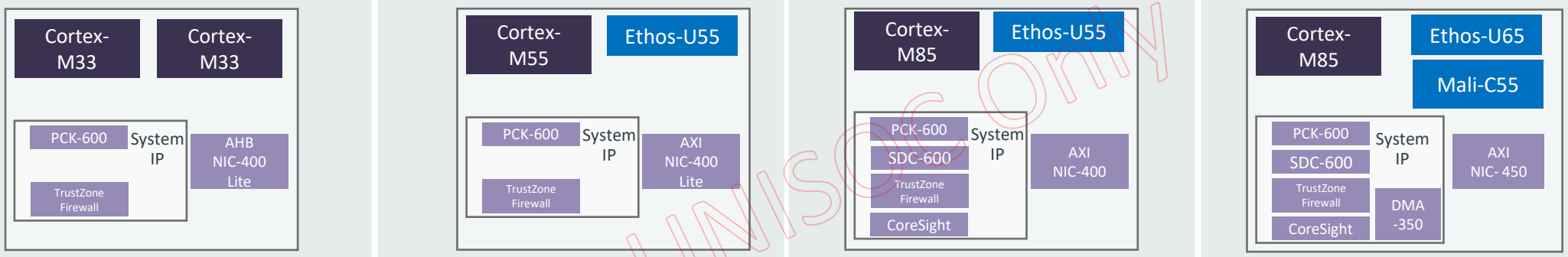
EndPoint AI solution for IoT

	Available			
Corstone	Corstone-201	Corstone-300	Corstone-310	Corstone-315
Main CPU	Cortex-M33	Cortex-M55	Cortex-M85 Cortex-M52	
NPU		Ethos-U55	Ethos-U65	Ethos-U85
ISP				Mali-C55
Solution Highlights	▪ **Armv8.0-M** with enhanced MPU and NVIC ▪ **TrustZone** hardware isolation for security ▪ **Arm Custom Instruction** (ACI) for acceleration of specific use cases ▪ Multi-core Debug and Power management	▪ **Cortex-M55** with **Armv8.1-M** ▪ **Helium** extension for enhanced CPU ML and DSP performance ▪ Heterogenous ML Compute with **Ethos-U55 NPU**	▪ **Cortex-M85** with highest scalar and floating-point performance **Pointer Authentication Codes** (PAC) and **Branch Target Identification** (BTI) for ROJ/JOP protection	▪ Integration of **Mali-C55 Image Signal Processor**, security IP and **DMA-350** ▪ Ready for **PSA Certified Level 2** ▪ Increase ML performance with **Ethos-U65 NPU**



Concept Adv. Planning Development Released

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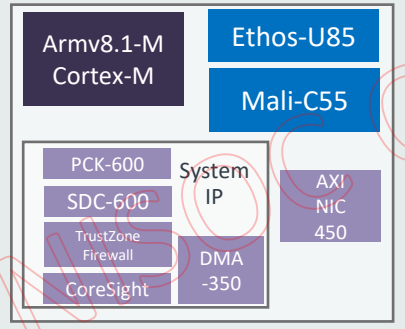
Corstone-320

	2024
Corstone	Corstone-320 
Main CPU	
NPU	Ethos-U85
ISP	
Solution Highlights	▪ **Any Armv8.1-M Cortex-M** with AXI interface ▪ **Transformer support** with Ethos **EthosU85** NPU ▪ Showcase Endpoint AI for Vision use cases with Transformer networks ▪ CPU/NPU choice ▪ Consolidates all Corstone-3xx into one Corstone-M

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Cortex-M85 performance, power and area (PPA)

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Cortex-M85 Scalar performance

	Cortex-M85 RTL	Cortex-M7 RTL	Cortex-M85 uplift
Dhrystone AC(DMIPS/MHz)	3.13	2.31	35.5%
Coremark AC (CM/MHz)	6.27	5.29	18.5%
Coremark IAR (CM/MHz)	5.79	4.92	17.7%

Conditions:

- Cortex-M7 r1p2 RTL
- Arm compiler 6.17, Armv8.1-M binaries on Cortex-M85
- Arm compiler 6.17, Armv7-M binaries on Cortex-M7
- IAR compiler 8.40.1, Armv7-M binaries

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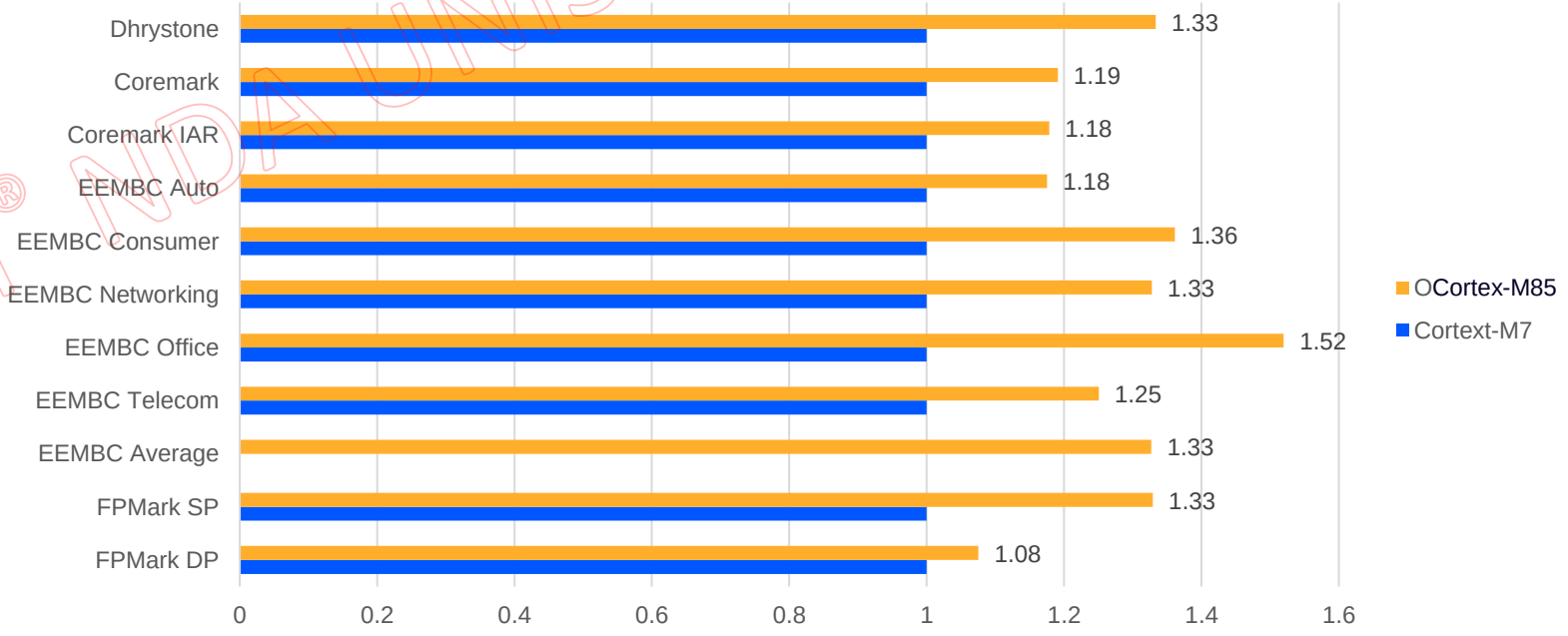

Conditions:

- Assumes processors running at the same clock frequency
- Code and data in 0 wait state TCM
- Benchmarks compiled with ARM tool-chain (v6.15)
- Cortex-M85: v8.1M binary, Cortex-M7: v7 binary
- EEMBC : EEMBCv1.1 suites

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Cortex-M85 Vector (MVE/Helium) Performance

Cortex-M85RTL vs Cortex-M55 RTL uplift

Class	count	q7	q15	q31	f16	f32
ActivationFunctions	6	21.7%	24.6%			
BasicMath	53	23.6%	30.6%	36.4%	29.8%	26.8%
ComplexMath	24		9.6%	7.2%	13.6%	12.3%
ControllerFunctions	15			12.0%		11.7%
FastMathFunctions	23		3.9%	5.1%	13.9%	8.9%
FilteringFunctions	81	21.1%	23.4%	21.2%	6.7%	11.1%
InterpolationFunctions	10	19.0%	17.7%	17.9%	4.5%	7.0%
MatrixFunctions	65	24.9%	27.9%	16.5%	18.3%	15.0%
Statistics	62	6.9%	5.5%	6.9%	10.5%	7.5%
SupportFunctions	45	19.9%	28.7%	26.9%	32.9%	25.5%
TransformFunctions	23		13.1%	10.8%	14.1%	8.2%

Not measured

>= 0%

>-10%, <0%

<-10%

Target Cycle Performance w.r.t Yamin

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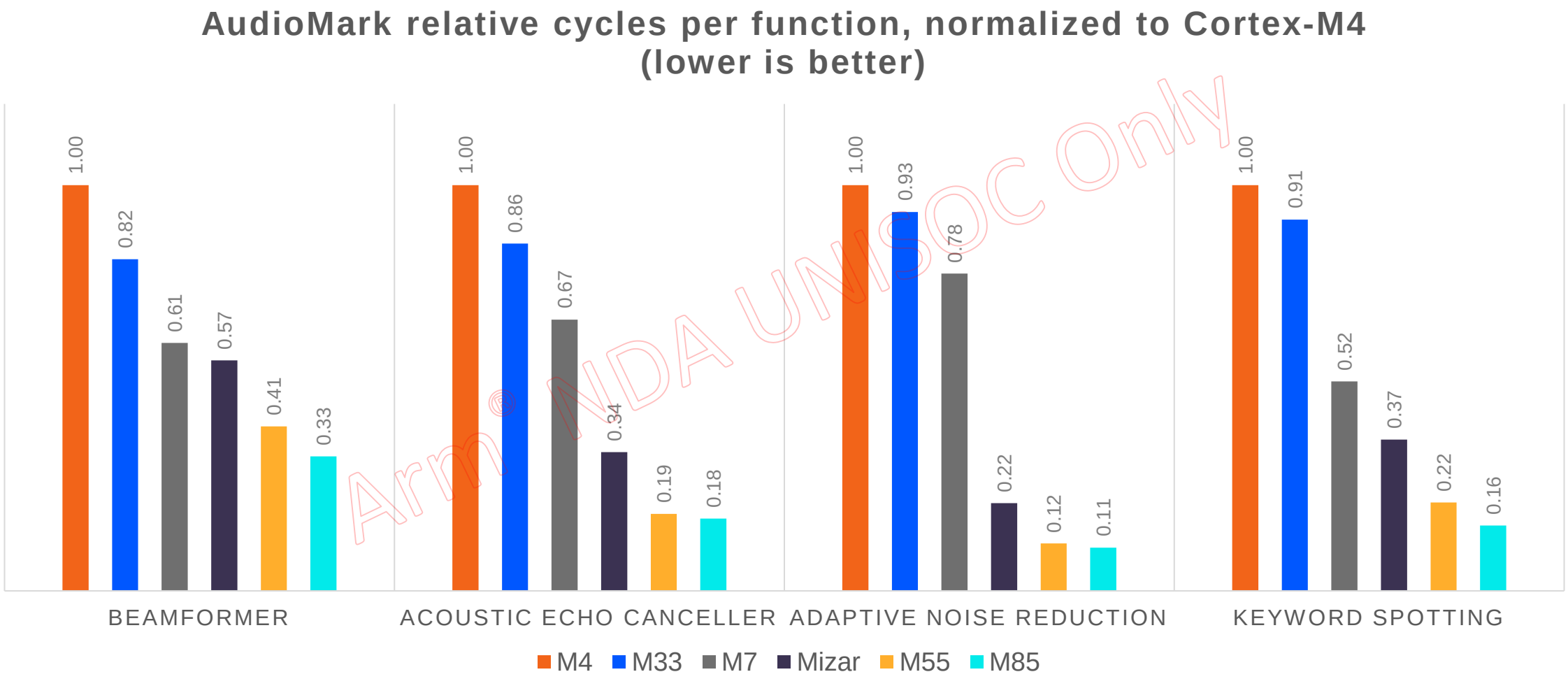
- Vector performance measured with CMSIS-DSP kernel libraries
- Iso-frequency comparison (cycle count comparison)
- Both Cortex-M55 and Cortex-M85 implement dual-beat Helium
- Performance uplift on Cortex-M85 due to microarchitecture improvements

AudioMark Performance Breakdown



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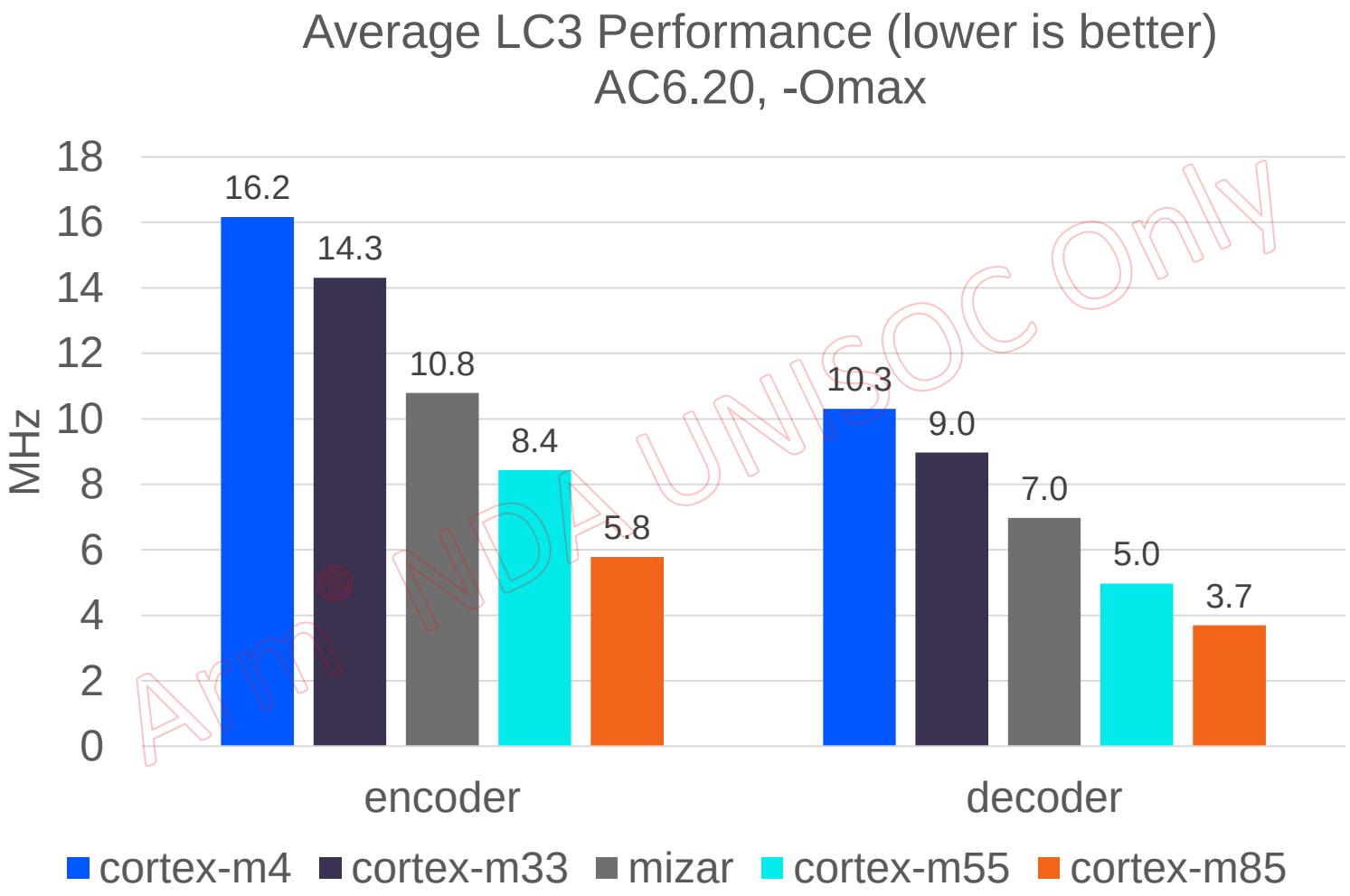
Note: Mizar is the project name for Cortex-M52

LC3 Performance

Espico



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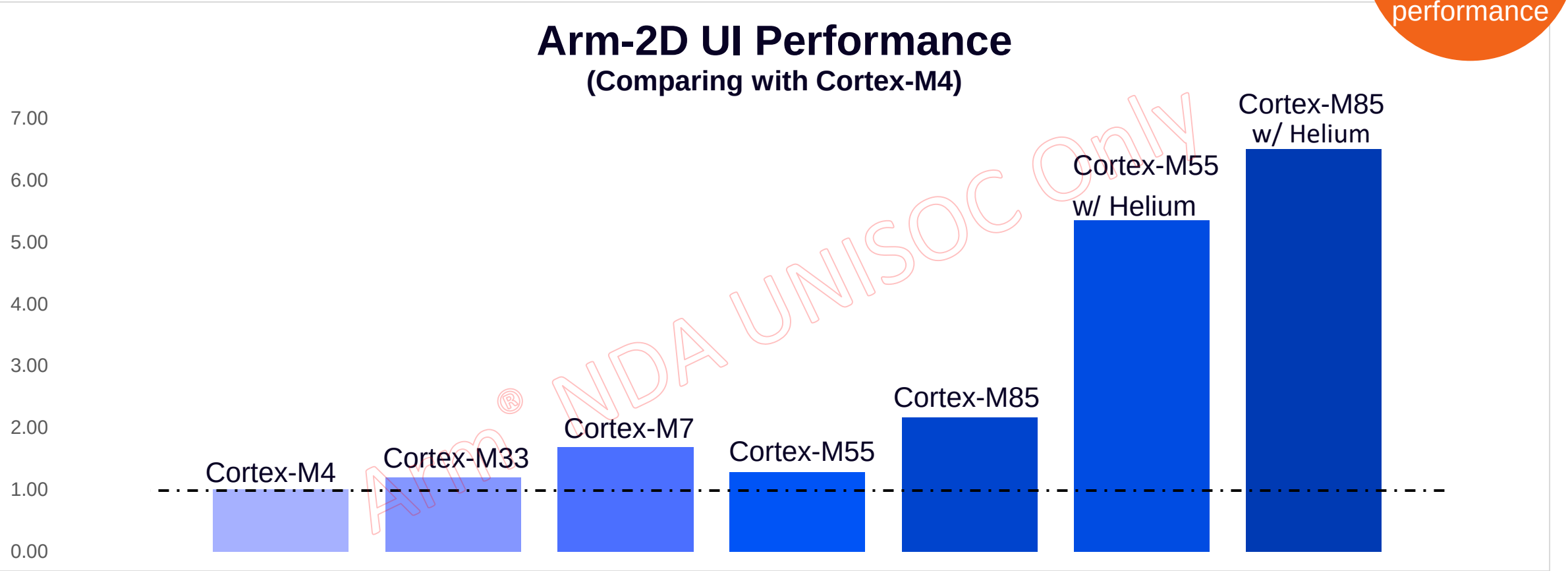
96kbps bitrate, 48kHz sample rate, 10msec frame size

Note: Mizar is the project name for Cortex-M52

Enhanced your User Experience with Helium

Up to **6x** UI performance



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NOTE：Cortex-M0+/M4/M33/M55/M85 Scalar numbers are generated from C code, the result can represent the average performance of existing embedded GUI software which written in C.
Cortex-M55 Helium is taking advantage of Helium instruction set (using assembly code and/or intrinsics), and Cortex-M55 Scalar uses the C implementation as reference.

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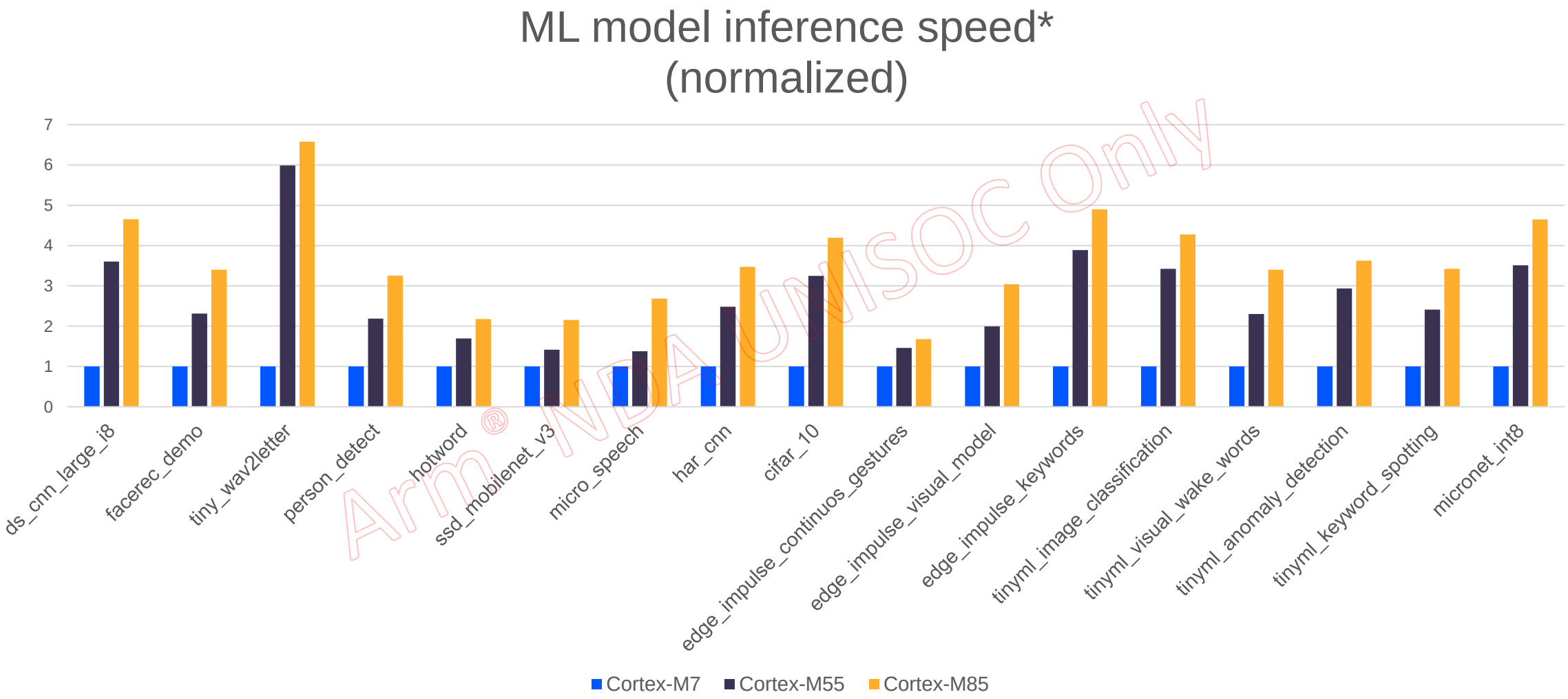
Neural network processing performance



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* execution from ideal TCM, iso-frequency

Power comparison

TSMC 22ULL	Cortex-M85 @600MHz	Cortex-M55 @600MHz
Dhrystone power*	95.9 mW/GHz	45.2 mW/GHz
Dhrystone efficiency	31.6 DMIPS/mW	35.4 DMIPS/mW

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Implementation library and corner
TSMC 22nm ttpg_0p9 sc7mcpp140z_cln22ul LVT

Processor configurations	RTL version	Configuration parameters
Cortex-M85	EAC	IWIC 1, IRQ 32, MVE 2, TZ 1, CPIF 1, MPU 8, MPU_S 8, SAU 4, ETM 1, ITM, 1, ECC 1, DCACHE 32K, ICACHE 32K, DBGLEVEL 2
Cortex-M55	REL	IWIC 1, IRQ 32, MVE 2, TZ 1, CPIF 1, MPU 8, MPU_S 8, SAU 4, ETM 1, ITM, 1, ECC 0, DCACHE 32K, ICACHE 32K, DBGLEVEL 2

*Dhrystone power includes cache RAM power

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Power comparison

Samsung 5LPE	Cortex-M85 @1.45GHz	Cortex-M7 @1.45GHz
Dhrystone power	54 mW/GHz	35 mW/GHz
Dhrystone efficiency	57 DMIPS/mW	63 DMIPS/mW

TSMC 22ULL	Cortex-M85 @1.1GHz	Cortex-M7 @1.1GHz
Dhrystone power	129 mW/GHz	86 mW/GHz
Dhrystone efficiency	23 DMIPS/mW	24 DMIPS/mW

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Implementation library and corner
Samsung 5lpe ttpg_0p75_m40__sigmax C8 LVT 7p5 track
TSMC 22nm ttpg_0p9 sc7mcpp140z_cln22ul LVT

Processor configurations	RTL version	Configuration parameters
Cortex-M85	EAC	IWIC 1, IRQ 32, MVE 2, TZ 1, CPIF 1, MPU 8, MPU_S 8, SAU 4, ETM 1, ITM, 1, ECC 1, DCACHE 32K, ICACHE 32K, DBGLEVEL 2
Cortex-M7	REL	WIC 1, IRQ 32, MPU 16, ETM 1, ITM, 1, ECC 1, DCACHE 32K, ICACHE 32K, DBGLEVEL 1

*Dhrystone power includes cache RAM power

22ULL floor plan area comparison

	Cortex-M85 no MVE, no FPU	Cortex-M7 no FPU	Increase over Cortex-M7	Cortex-M85 w/ FPU (no MVE)	Cortex-M85 w/ MVE, w/ FPU	Cortex-M7 w/ FPU	Increase over Cortex-M7
1.1GHz floor plan area*	0.879 mm^2	0.571 mm^2	53%		1.09 mm^2	0.617 mm^2	77%
600 MHz floor plan area*				0.858 mm^2		0.575 mm^2	49%
300MHz floor plan area*	0.747 mm^2	0.533 mm^2	40%		0.885 mm^2	0.566 mm^2	56%

Process conditions	22ULL
Process	TSMC CLN22ULL
Options	7 Track LVT
Slow corner	sspg_0p81v_sc7mcpp140z

Processor configurations	RTL version	Configuration parameters
Cortex-M85	EAC	IWIC 1, IRQ 32, MVE 2, TZ 1, CPIF 1, MPU 8, MPU_S 8, SAU 4, ETM 1, ITM, 1, ECC 1, DCACHE 32K, ICACHE 32K, DBGLEVEL 2
Cortex-M7	REL	WIC 1, IRQ 32, MPU 16, ETM 1, ITM, 1, ECC 1, DCACHE 32K, ICACHE 32K, DBGLEVEL 1

* Includes cache RAM area

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5LPE floor plan area comparison

	Cortex-M85 no MVE, no FPU	Cortex-M7 no FPU	Increase over Cortex-M7	Cortex-M85 w/ MVE, w/ FPU	Cortex-M7 w/ FPU	Increase over Cortex-M7
1.5GHz floor plan area*	0.135 mm^2	0.0828 mm^2	63%	0.168 mm^2	0.0907 mm^2	85%
500MHz floor plan area*	0.111 mm^2	0.0765 mm^2	45%	0.134 mm^2	0.084 mm^2	60%

Process conditions	5LPE
Process	Samsung 5LPE
Options	7p5 Track LVT
Slow corner	sspg_0p675_m40_tc

Processor configurations	RTL version	Configuration parameters
Cortex-M85	EAC	IWIC 1, IRQ 32, MVE 2, TZ 1, CPIF 1, MPU 8, MPU_S 8, SAU 4, ETM 1, ITM, 1, ECC 1, DCACHE 32K, ICACHE 32K, DBGLEVEL 2
Cortex-M7	REL	WIC 1, IRQ 32, MPU 16, ETM 1, ITM, 1, ECC 1, DCACHE 32K, ICACHE 32K, DBGLEVEL 1

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* Includes cache RAM area

Cortex-M85 vs. Cortex-M7 Enhancements

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Cortex-M85 Capability Enhancements (1/3)

Cortex-M7	Cortex-M85	Comments
Legacy DSP	Legacy DSP + optional Helium	Helium can enable better performance in normal data processing via auto-vectorisation support in C compilers.
	Low overhead branch extension	Increase performance without unrolling loops (reduce code size)
	TrustZone	Enable security and firmware protection.
	PACBTI	Additional security extension to mitigate against RoP/JoP attacks
	Unprivileged Debug Extension (UDE)	Allows 3rd party unprivileged firmware to be developed without exposing secrets.
MPU: 0/8/16 MPU regions, PMSAv7	MPU: 0/4/8/12/16 MPU regions, PMSAv8	MPU regions in Armv8-M are easier to setup. New PXN memory attribute (Armv8.1-M) to prevent unprivileged code to execute at privileged level (escalation attack).
	Performance Monitoring Unit (PMU)	Very helpful in checking cache hit/miss, alone with other performance monitor uses
Optional single precision, or single + double precision FPU	Optional half+single+double precision FPU	Olympus adds half precision support

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Cortex-M85 Capability Enhancements (2/3)

Cortex-M7	Cortex-M85	Comments
TCM ECC require custom ECC logic outside of the processor	TCM ECC built-in	Easier system integration
Up to 240 interrupts	Up to 480 interrupts. Each IRQ line can be disabled at synthesis.	Olympus has better Interrupt configurability.
Up to 4 DWT comparators	Up to 8 DWT comparators	DWT comparators are multiple functions (Data watchpoint, data trace, ETM trigger) and 4 is not quite enough.
ETM: instruction trace, and optional full data trace	ETM: instruction trace only	Full data trace is rarely needed. DWT supports selective data trace so you can still trace some data without full data trace support.
	Coprocessor interface	Custom defined acceleration solution or use as a low latency access path to control or I/O registers.
	Arm Custom Instructions (planned for Olympus r1 in 2022)	Custom defined acceleration solution
FPU, debug (excluding ETM) and core logic has single power domains	Separated power domains for core, EPU, debug	Better power control in Olympus. Cache SRAM is separated power domain for both.

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Cortex-M85 Capability Enhancements (3/3)

Cortex-M7	Cortex-M85	Comments
Internal WIC only	Internal + External WIC support	External WIC is useful for handling wakeup control in an asynchronous clock domain.
Single control bit for D-cache enable/disable of lookup and allocate - Disabling D-cache might require stopping interrupts	Disabling D-cache allocate does not immediately stop cache look up. C friendly and no complication of D-cache data during cache disabling.	Easier cache management.
32-bit AHB slave port	64-bit AHB slave port with burst support	DMA controller accessing TCMs can have a much higher data bandwidth.
Online MBIST integration need to be handled by SoC designer	Internal PCM-100 for online MBIST operations	A useful FuSa feature.
	Optional Bus Interface protection (parity)	Can be important for FuSa
Lock signals for TCM and PPB control registers, VTOR	Lock signals for TCM and PPB control registers, VTOR, MPUs, SAU, AIRCR	Olympus lock signal is a super set of Cortex-M7
ECC error bank registers	ECC error bank registers + RAS extension	ECC error reporting capabilities are similar between the two processors. As a part of RAS extension, Olympus supports poison signals on its AXI interface.

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Cortex-M85 micro-architecture

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Processor design

- In-order 7-stage scalar pipeline with dual-issue and instruction fusion
  - 9-10-stage Vector and floating point pipeline
  - Harvard instruction fetch and data access
- Integrated memory system with support for Tightly Coupled Memory (TCMs) and Level 1 Harvard instruction and data cache
- Direct, high bandwidth DMA access to TCM interfaces using an integrated AMBA5 AHB5 slave interface.

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Interfaces

- 64-bit AMBA5 AXI5 master interface for efficient access to high-latency peripherals such as SRAM, DDR and Flash memory
- Single 64-bit instruction TCM interface mapped to CODE memory region
- Four 32-bit quad banked Data TCM interfaces mapped to SRAM memory region supporting efficient, high bandwidth access from both the processor and DMA to local low-latency memory
- 64-bit AMBA5 AHB5 slave interface for system access to TCM
- Tightly coupled 32-bit AMBA5 AHB5 interface for efficient access to system peripherals
- 32-bit AMBA 4 APB External Peripheral Port, EPPB interface suitable for connection to external debug peripherals
- System-level TrustZone support on all memory interfaces including internal security gating on TCM interfaces compliant with the ARM Trusted Base System Architecture (TBSA) for ARMv8-M
- External IDAU interface to allow security attributes to be defined by the system
- 8-bit AMBA 4 ATB interfaces for ITM and ETM trace
- 32-bit AMBA AHB slave interface for debug access to registers, memory and peripherals
- Optional CoreSight Cross Trigger Interface, CTI, for debug and trace synchronization

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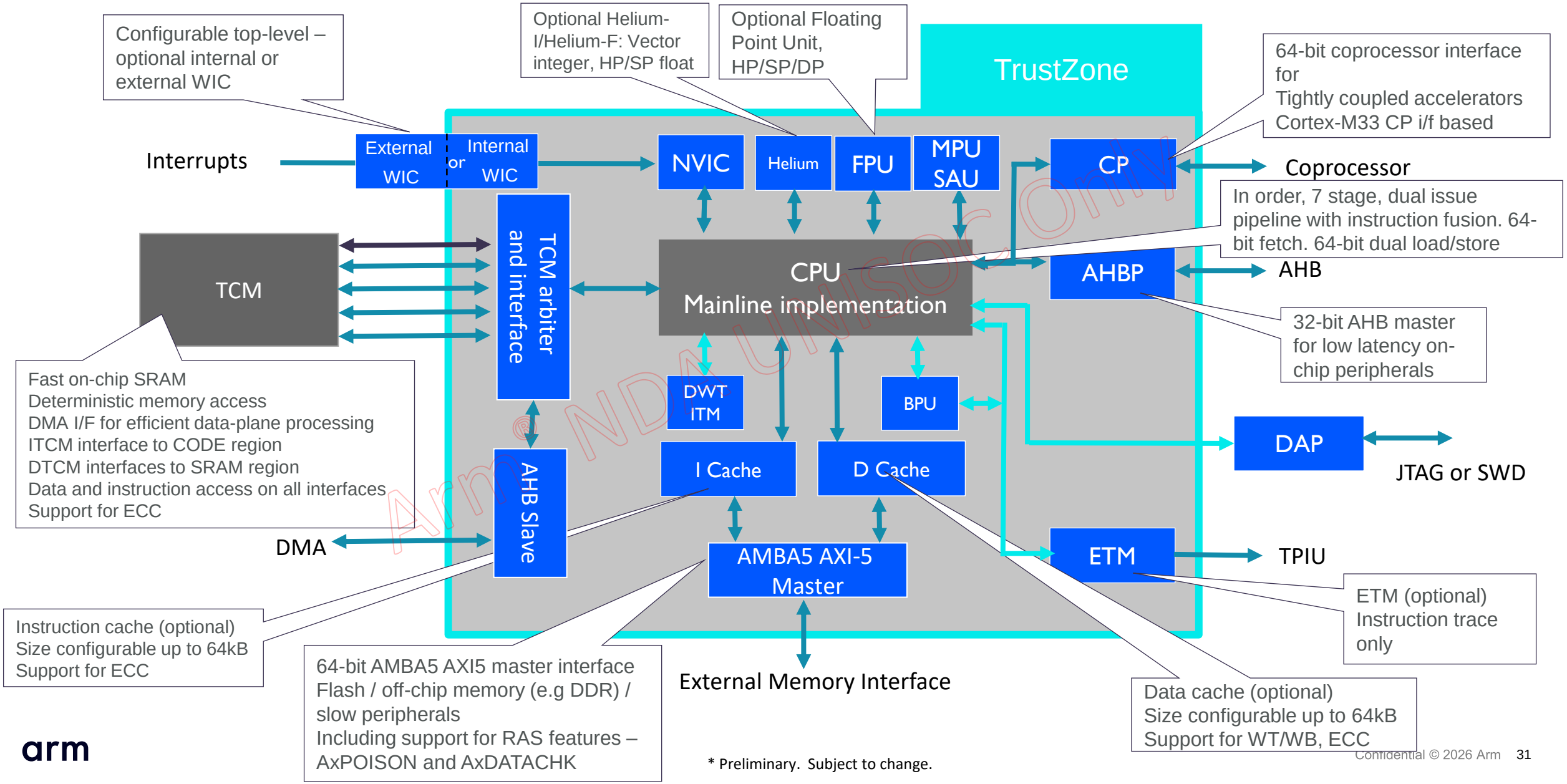
Block diagram of Cortex-M85



* Preliminary. Subject to change.

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Functional safety and reliability

- L1 Cache and TCM interfaces support optional internal Error checking and correction (ECC) – single bit correct/ double bit detect
- All ECC errors reported to the system on an external interface
- Support for Dual core lock step processor operation
- On-line MBIST – support for memory integrity checking during processor runtime

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Power aware

- Support for multiple power-domain SRPG implementation through UPF
- ARM standard P-Channel and Q-Channel based power control
- Support for system level and internal Wake-up Interrupt controller (WIC)

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Cortex-M85 value proposition

Performance

- **Gain new markets with the highest performance Cortex-M, ever!**
  - Attain performance levels previously only achievable with complex heterogeneous systems
  - Reduce cost and system complexity for high performance systems with a single Cortex-M processor

Security

- **Empower developers with a robust and simple-to-use architecture with built-in security**
  - Leverage the open-source Trusted firmware-M for Armv8-M architectures
  - Take advantage of the largest ecosystem of software, including optimized DSP and NN libraries, available to embedded system designers

Innovation

- **Rapidly capitalize on emerging market opportunities with Helium™**
  - Explore new machine learning use-cases at the device edge
  - Implement multi-channel high fidelity audio on DSP-less microcontrollers
  - Drive down BOM cost for robots and industrial control systems

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Tack
ಧನ್ಯವಾದಗಳು
Merci
Danke
Gracias
Grazie
谢谢
ありがとう
Asante
Thank you
감사합니다
धन्यवाद
Kiitos
شكرًا
ধন্যবাদ
תודה
ధన్యవాదములు
Köszönöm

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